Equations of parallel and perpendicu[redacted]nes

1

a They have equal slopes of $m = 2$.

b

i No

ii No

iii Yes

c Equations with the same slope (m) have graphs that have the same angle of inclination.

2

a Yes

b No

c No

3

a Yes

b No

c Yes

d No

e Yes

f No

4

a -3

b $\frac{4}{5}$

5 $y = -2, y = -6$

6 $y = 3x + 4$

7

a $y = 5$

b $x = -7$

c $y = -\frac{7x}{6} - 2$

d $y = -8x + 7$

e $y = 2x + 1$

f $y = \frac{2}{3}x - 1$

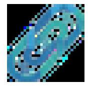
8 $4x - 3y = 7$

9

a $m = -\frac{13}{4}$

b $y = -\frac{13x}{4} - \frac{5}{4}$

a $m = 13$

 $y = 13x + 5$

11 $m = -\frac{4}{5}$

12

a

i $m = 4$

ii $b = -32$

iii $y = 4x - 32$ or $4x - y = 32$

b

i $m = -1$

ii $b = 0$

iii $y = -x$ or $x + y = 0$

c

i $m = 5$

ii $b = 26$

iii $y = 5x + 26$ or $5x - y = -26$

13

a $-\frac{2}{5}$

b $\frac{9}{10}$

c $-\frac{1}{3}$

d 4

14 Two lines that intersect each other at an angle of 90°.

15 -1

16

a No

b No

c Yes

d Yes

e Yes

f No

g Yes

17

a $\frac{1}{9}$

b 3

c $\frac{1}{5}$

d 1

18

a $-\frac{1}{10}$

$-\frac{4}{3}$

c $\frac{1}{10}$

d $-\frac{5}{2}$

19

a No

b Yes

c Yes

d No

e No

20

a No

b No

c No

d Yes

21

a $x=10$

b $y=-7$

c $x+7y=6$

d $4x+7y=-12$

e $7x+2y=-31$

22

a $m=\frac{3}{20}$

b $y=\frac{3x}{20}-\frac{14}{5}$ or $3x-20y=56$

23

a $y=2$

b $3x+4y=8$

24

a $8x+3y=14$

b Yes

25

a $2x-5y=8$

b No

26

a $m=-\frac{1}{5}$

b $y=-\frac{x}{5}+6$

27 $y=-\frac{1}{10}x+3$

Additional questions

28

a 2

b $-\frac{7}{5}$

c $\frac{1}{2}$

d $-\frac{2}{3}$

29

a Perpendicular

b Neither

c Parallel

30

a If two lines are parallel, their slopes will be equal.

b If two lines are perpendicular, their slopes will be opposite reciprocals of each other. This means that the product of their slopes will be -1.

31

a $m = 4$

b $m = -\frac{1}{3}$

c $m = \frac{1}{9}$

d $m = \frac{1}{5}$

e $m = 1$

32

a $y = \frac{2}{3}x + 1$ is parallel

b $4x - 6y = 37$ is parallel

c $3x + 2y = -13$ is not parallel

d $y = -\frac{2}{3}x - 8$ is not parallel

33

a $y = \frac{x}{4} - \frac{3}{2}$ is not perpendicular

b $4x + 16y = 96$ is perpendicular

c $x + 4y = -3$ is perpendicular

d $y = -4x - 3$ is not perpendicular

34

a Parallel

b Parallel

c Perpendicular

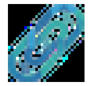
d Neither

35

a $y = -\frac{7x}{\ } - 2$

b $x = -4$

c $y=5$

 $x=-7$

36

a $y=1$

b $y=-\frac{1}{2}x+4$

c $x=10$

d $y=-7$

37

a $m=-\frac{13}{4}$

b $y=-\frac{13}{4}x-\frac{5}{4}$

38

a $m=13$

b $y=-\frac{1}{13}x+5$

39

a $y=\frac{3}{4}x-4$

b $-\frac{4}{3}$

c $4x+3y=12$

40 $m=10$

41 If we calculate the slopes of the two lines, we get $-\frac{b}{a}$ and $\frac{a}{b}$ for the two lines. Since these slopes are opposite reciprocals of each other, the two lines must be perpendicular.

42 Yes. Since Ammon and Minh earn money at the same rate, we can expect the graphs of their bank balances to have the same slope. Therefore, the two lines will be parallel.

43 Since the lines are perpendicular, the product of their slopes will be $\frac{p}{4}\times\frac{q}{-5}=-1$. This tells us that $pq=20$.